

Project 4: Medical research AI Assistant with Graph RAG & Vector DB

Concept

Construct a sophisticated knowledge graph integrating data from various biomedical domains: genes, proteins, diseases, drugs, pathways, symptoms, clinical trials, and scientific literature.

The RAG component would then be used to query this KG to generate novel hypotheses for disease mechanisms, identify potential drug targets, or suggest existing drugs for new therapeutic uses (drug repurposing).

Healthcare Value

- **Accelerated Drug Discovery:** Reduce the time and cost associated with identifying new drug candidates.
- **New Therapeutic Avenues:** Discover unexpected connections between existing drugs and diseases.
- **Enhanced Research Efficiency:** Streamline the process of hypothesis generation for medical researchers.

Key Features for Demo

- **Nested Queries:**
 - "Which drugs are known to interact with Protein X (which is implicated in Disease Y) and have shown positive outcomes in clinical trials for conditions with similar inflammatory pathways to Disease Y?" (Multi-hop query spanning proteins, diseases, drugs, and clinical trial data).
 - "Identify all genes that are upregulated in both Condition A and Condition B, and for which there are no known drugs currently targeting their protein products, but which are linked to a metabolic pathway affected by Drug Z."
- **Discovery of Not-So-Easy-to-Find Patterns:**
 - Identify potential drug repurposing candidates by discovering indirect relationships (e.g., Drug A, approved for Condition X, affects a protein that interacts with another protein implicated in Condition Y, for which there is no effective treatment).

- Uncover subtle co-occurrences of genes/proteins and specific disease phenotypes in scientific literature that haven't been explicitly studied together but suggest a novel therapeutic target.
- Detect unexpected adverse events or off-target effects of drugs by looking for subtle correlations between drug usage patterns and reported symptoms/conditions across patient cohorts (synthetically generated for privacy).

Suggested approach

"Which drugs are known to interact with **Protein X** (which is implicated in **Disease Y**) and have shown positive outcomes in clinical trials for conditions with similar **inflammatory pathways** to Disease Y?"

- "Protein X," "Disease Y," "inflammatory pathways" (semantic search on research abstracts, or direct KG query for entities).
- "drugs known to interact," "positive outcomes in clinical trials" (KG traversal for relationships, and potentially semantic search on clinical trial summaries for nuanced outcomes).

Hybrid: Initial semantic search on literature to identify relevant proteins/diseases, then use these as entry points into the KG to explore complex relationships (protein-disease, drug-protein, disease-pathway, clinical trial results).

Knowledge Graph (KG) forms the backbone for representing structured biomedical entities (drugs, genes, proteins, diseases, pathways, symptoms, mechanisms) and their explicit relationships, a **Vector Database (VectorDB)** plays a crucial and complementary role, especially in the context of "Hypothesis Generation" and leveraging vast amounts of unstructured scientific data.

Data Sources:

- 1) Biomedical entities and relations → will go in neo4j
- 2) relevant medical research papers → will go in vector DB

Synthetic:

Use Claude or another AI tool to create a python script that will generate meaningful data for above.

In the prompt, you can copy paste the above details and ask it to create a python script that generates the data and contains the patterns in the data to meaningfully answer the questions mentioned above. You can also ask it to put the pattern validation code in the script!

The same approach can be used to create synthetic research papers relevant to the questions and use-cases above.

A generated synthetic dataset is provided in the project-4-dataset folder

=== DATA GENERATION SUMMARY ===

Genes: 100

Proteins: 80

Diseases: 60

Drugs: 70

Pathways: 41

Symptoms: 12

Clinical Trials: 50

Literature Papers: 100

Relationships:

gene_protein: 121

protein_disease: 226

drug_protein: 148

drug_disease: 134

pathway_gene: 366

pathway_protein: 274

disease_symptom: 240

clinical_trial_drug: 84

clinical_trial_disease: 50

protein_protein_interaction: 100

drug_adverse_events: 189

Total Relationships: 1932

=== VALIDATION REPORT ===

1. Multi-hop query validation:

- ✓ PASS: Drugs with positive trials that target disease-associated proteins: 25
- ✓ PASS: Multi-hop examples (Drug->Protein->Disease): 25
- ✓ PASS: Inflammatory pathways found: 3

2. Co-upregulated genes validation:

- ✓ PASS: Co-upregulated genes in both conditions: 8
- ✓ PASS: Untargeted co-upregulated genes: 8
- ✓ PASS: Genes in metabolic pathways: 32
- ✓ PASS: Drug Z affects metabolic pathways: 1 connections

3. Drug repurposing validation:

- ✓ PASS: Diseases without effective treatment: 1
- ✓ PASS: Potential repurposing candidates (indirect connections): 18
- ✓ PASS: Specific repurposing examples found: 18

4. Adverse events validation:

- ✓ PASS: Drugs with adverse events: 70
- ✓ PASS: Unique adverse event types: 12
- ✓ PASS: Rare adverse events (< 5% frequency): 28

5. Literature patterns validation:

- ✓ PASS: Gene-protein relationships: 121
- ✓ PASS: Protein-disease relationships: 226
- ✓ PASS: Pathway coverage: 41/41 (100.0%)

6. Demo query specific patterns:

- ✓ PASS: Strong protein-disease implications: 54
- ✓ PASS: Successful trials in inflammatory conditions: 2

=== OVERALL VALIDATION RESULT ===

✓ ALL PATTERNS VALIDATED SUCCESSFULLY

✓ Data is ready for demo queries

Data saved to medical_knowledge_graph_data.json

=== SCRIPT EXECUTION COMPLETE ===

✓ All patterns validated successfully

✓ Data is ready for RAG-based knowledge graph queries

✓ File saved: medical_knowledge_graph_data.json

Medical research paper pdfs also provided

Demo queries that can now be answered:

1. Multi-hop: Which drugs interact with proteins implicated in inflammatory diseases and have positive clinical trial outcomes?
2. Pattern discovery: Find genes upregulated in multiple conditions with no targeting drugs but linked to metabolic pathways.
3. Drug repurposing: Identify drugs approved for one condition that could treat another via protein-protein interactions.
4. Adverse events: Detect subtle correlations between drug usage and rare symptoms.
5. Literature mining: Discover novel therapeutic targets through co-occurrence patterns.